

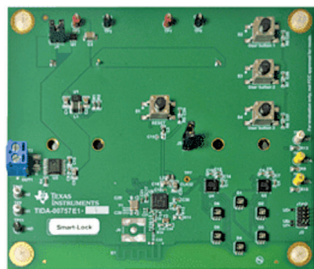
appliance energy use, or in connected home appliances like air-conditioners, refrigerators, or heaters for remote management.

Link. <https://www.electronics-foru.com/electronics-projects/industry-grade-projects/reference-design-for-wireless-energy-monitor-system>

OEM Brand. Microchip Technology

Smart lock

Wireless, battery-powered smart locks are becoming increasingly popular as building owners and homeowners retrofit doors with electronic locking systems. The key design challenge for engineers is balancing functionality with battery



life. High-current motors and radios can quickly drain batteries, and replacing them frequently is time-consuming and costly, which makes low average current consumption a primary design goal. This reference design demonstrates how engineers can extend battery life using a highly efficient power topology. The design enables wireless locking and unlocking of doors, battery voltage monitoring for low-life alerts, motor current limiting to extend motor life, and LED-based event communication.

Engineers can calculate average power consumption and estimate battery life by focusing on the main devices in a smart lock, which include the MCU with integrated BLE, motor driver, and power management components.

Applications. This reference design can be used for smart lock systems and other battery-powered applications.

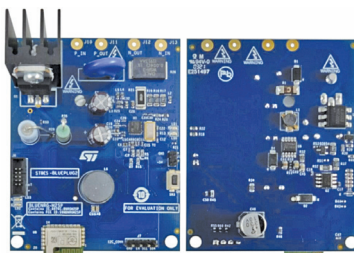
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[icsforu.com/electronics-projects/smart-lock-reference-design-2](https://www.electronics-foru.com/electronics-projects/smart-lock-reference-design-2)

OEM Brand. Texas Instruments (TI)

Smart plug

Smart devices that control electronics remotely are common, and this smart plug reference design simplifies home automation for engineers. It uses Bluetooth low energy (BLE)



5.2 to securely transmit metering data from electrical loads to a BLE-supported smartphone, functioning as both BLE master and slave. Users can manage loads via a smartphone app or other smart devices, including turning devices on or off, scheduling, dimming, and tracking energy use. The plug consumes 3-7mA on AC mains, features a metering chip for power and energy measurement via shunt sensors, a three-terminal TRIAC for AC switching, and a non-isolated buck converter.

The BLE module handles app communication, while the metering IC monitors supply and loads. The board also supports EEPROM data storage, includes a manual switch for direct control, and has LEDs indicating connectivity status.

Applications. The reference design is suitable for applications requiring a compact form factor and low current draw.

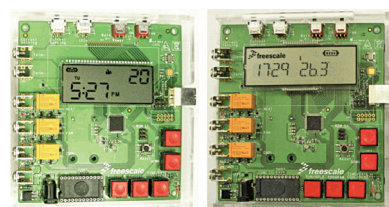
Link. <https://www.electronics-foru.com/electronics-projects/reference-design-of-a-smart-plug-for-home-automation>

OEM Brand. STMicroelectronics

Thermostat

When designing a thermostat or smart-home relay node, engineers face challenges such as maintaining

low idle power, interfacing directly with HVAC systems, using displays that do not wake the MCU, and ensuring firmware stability during hardware swaps. This reference design demonstrates how to create a low-power thermostat or smart-home relay. The board uses latching relays to control HVAC systems, which consume power only when switching, and connect to standard 24V AC wires. It measures temperature using a thermistor and resistor, can run on 6-12V DC, and includes a programming connector.



This design is versatile enough for thermostats, relay nodes, or other smart-home devices that mostly remain in sleep mode, wake on events, and display information efficiently. It can also serve as a base for devices such as blinds, valves, or room controllers with front displays.

Applications. This reference design can be used for thermostats, smart-home relays, motorised blinds or shades, water or gas valves, and room controllers with displays.

Link. <https://www.electronics-foru.com/electronics-projects/thermostat-reference-design>

OEM Brand. Freescale Semiconductor

The reference designs discussed above help you make everyday devices smarter, more efficient, and easier to control. They allow better power management, energy tracking, and automation across a range of devices. These are valuable resources for designing smart plugs, locks, thermostats, and other connected home devices. **EFY**

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